

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I Athavan K(192524036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student :Athavan k

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough → Possible Flu
- li. If a patient has Fever AND Rash → Possible Measles
- lii. If a patient has Cough AND Breathlessness → Possible Pneumonia
- Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) $P1: \text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) $P2: \text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) $P3: \neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

- (a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.
- (b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.
- (c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Symbol definitions:

- F = Patient has Fever
- C = Patient has Cough
- R = Patient has Rash
- B = Patient has Breathlessness
- M = Possible Flu
- Me = Possible Measles
- Pn = Possible Pneumonia

Knowledge base rules:

Rule 1: $F \wedge C \rightarrow M$ (Fever AND Cough $\rightarrow$ Possible Flu)

Rule 2: $F \wedge R \rightarrow Me$ (Fever AND Rash $\rightarrow$ Possible Measles)

Rule 3: $C \wedge B \rightarrow Pn$ (Cough AND Breathlessness $\rightarrow$ Possible Pneumonia)

Facts about Patient A:

$F = \text{True}$, $C = \text{True}$, $B = \text{True}$ (R is not asserted, so $R = \text{False/unknown}$)

(b) Forward Chaining

Forward chaining starts from known facts and fires rules whose premises are fully satisfied, adding new derived facts until no more rules can fire.

Step	Known Facts (Agenda)	Rule Checked	Premise Satisfied?	Fact Derived
1	F, C, B	R1: $F \wedge C \rightarrow M$	$F=T, C=T \rightarrow \text{Yes}$	M (Possible Flu) added
2	F, C, B, M	R2: $F \wedge R \rightarrow Me$	$F=T, R=\text{unknown} \rightarrow \text{No}$	No new fact (R not established)
3	F, C, B, M	R3: $C \wedge B \rightarrow Pn$	$C=T, B=T \rightarrow \text{Yes}$	Pn (Possible Pneumonia) added
4	F, C, B, M, Pn	All rules re-checked	No further rule satisfiable	Fixpoint reached — stop

Final derived diagnoses for Patient A: Possible Flu (M) and Possible Pneumonia (Pn). Possible Measles is not established because Rash (R) was never asserted as true.

(c) Backward Chaining — Goal: 'Patient A has Pneumonia'

Backward chaining starts from the goal and recursively reduces it to sub-goals until each sub-goal matches a known fact (a leaf in the AND/OR goal-reduction tree).

Goal: P_n

P_n is concluded by Rule 3: $C \wedge B \rightarrow P_n$

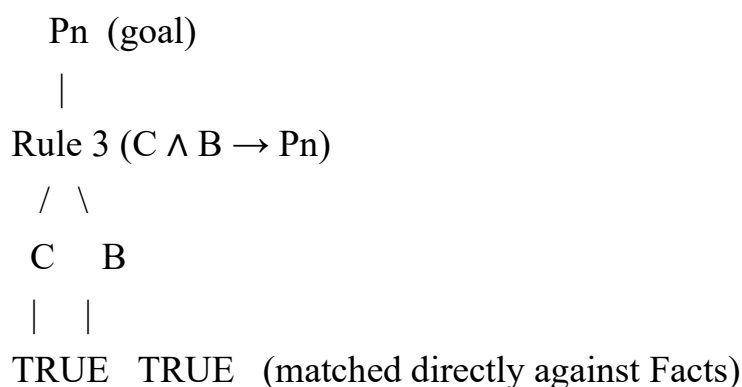
So P_n reduces to the AND of two sub-goals: $C \text{ AND } B$

Sub-goal C (Cough): matches Fact ' $C = \text{True}$ ' $\rightarrow$ SATISFIED (leaf)

Sub-goal B (Breathlessness): matches Fact ' $B = \text{True}$ ' $\rightarrow$ SATISFIED (leaf)

Both sub-goals are True $\Rightarrow P_n$ is proved TRUE

Goal Reduction Tree:



Conclusion: Since both leaf sub-goals C and B are already known facts, the AND node succeeds, and the goal 'Patient A has Pneumonia' is formally verified as TRUE using backward chaining.

Case Study 2 — Autonomous Traffic Management Agent

(a) Unification of Rule 1 with the Facts

Rule 1: $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Facts available: $\text{Vehicle}(\text{Ambulance})$, $\text{EmergencyType}(\text{Ambulance})$

Unification proceeds literal by literal, building up the Most General Unifier (MGU) θ :

Step	Expression Pair Being Unified	Substitution Applied	Running θ
1	$\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance})$	$x / \text{Ambulance}$	$\theta = \{x/\text{Ambulance}\}$
2	$\text{EmergencyType}(x)$ with $\text{EmergencyType}(\text{Ambulance})$	x already bound to Ambulance — consistent, no conflict	$\theta = \{x/\text{Ambulance}\}$
3	Apply θ to conclusion $\text{ClearPath}(x)$	$x/\text{Ambulance}$ substituted	Result: $\text{ClearPath}(\text{Ambulance})$

Final MGU: $\theta = \{x/\text{Ambulance}\}$. Applying θ to Rule 1's conclusion yields the new fact $\text{ClearPath}(\text{Ambulance})$.

(b) Resolution Proof that $\text{Proceed}(\text{CarA})$ is True

Step 1 — Convert each rule and the negated goal to CNF (clause form):

C1 (Rule1): $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

C2 (Rule2): $\neg \text{ClearPath}(y) \vee \text{GreenSignal}(y)$

C3 (Rule3): $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

C4 (Fact): $\text{Vehicle}(\text{Ambulance})$

C5 (Fact): $\text{EmergencyType}(\text{Ambulance})$

C6 (Fact): $\text{Vehicle}(\text{CarA})$

C7 (Fact): $\text{Behind}(\text{CarA}, \text{Ambulance})$

C8 (Negated goal): $\neg \text{Proceed}(\text{CarA})$

Step 2 — Resolution steps:

#	Resolve	Unifier	Resulting Clause
1	C1 with C4 ($\text{Vehicle}(\text{Ambulance})$)	$\{x/\text{Ambulance}\}$	$\neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$
2	Result-1 with C5 ($\text{EmergencyType}(\text{Ambulance})$)	—	$\text{ClearPath}(\text{Ambulance})$
3	C2 with Result-2 ($\text{ClearPath}(\text{Ambulance})$)	$\{y/\text{Ambulance}\}$	$\text{GreenSignal}(\text{Ambulance})$

#	Resolve	Unifier	Resulting Clause
4	C3 with C6 (Vehicle(CarA))	{x/CarA}	$\neg \text{Behind}(\text{CarA}, y) \vee$ $\neg \text{GreenSignal}(y) \vee$ $\text{Proceed}(\text{CarA})$
5	Result-4 with C7 (Behind(CarA, Ambulance))	{y/Ambulance}	$\neg \text{GreenSignal}(\text{Ambulance}) \vee$ $\text{Proceed}(\text{CarA})$
6	Result-5 with Result-3 (GreenSignal(Ambulance))	—	$\text{Proceed}(\text{CarA})$
7	Result-6 with C8 ($\neg \text{Proceed}(\text{CarA})$)	—	$\square$ (empty clause — contradiction)

Since resolving the KB together with the negated goal produces the empty clause $\square$, the original goal $\text{Proceed}(\text{CarA})$ is proved TRUE by refutation.

(c) Sufficiency Analysis for Two Simultaneous Emergency Vehicles

The current KB is NOT sufficient to handle two simultaneous emergency vehicles.

Reasoning:

- Rule 1 independently grants $\text{ClearPath}(x)$ to every vehicle that is $\text{Vehicle}(x) \wedge \text{EmergencyType}(x)$. If two ambulances/emergency vehicles approach the same intersection from conflicting directions, both would simultaneously satisfy Rule 1 and both would receive GreenSignal via Rule 2, which is physically unsafe (a signal conflict).
- There is no predicate capturing conflict or right-of-way, e.g. $\text{Conflict}(x,y)$ or $\text{IntersectionOf}(x,y)$, so the KB cannot detect that two emergency vehicles occupy competing paths.
- There is no priority ordering among emergency vehicles (e.g., based on arrival time, severity, or vehicle type such as ambulance vs fire truck).

Additional rules/facts needed:

- $\text{Priority}(x,y)$ — a precedence fact/relation between two emergency vehicles, e.g., derived from arrival time or dispatch priority.
- $\text{Conflict}(x,y)$ — asserts that x and y cannot both proceed simultaneously (e.g., they cross the same junction).
- A refined arbitration rule: $\forall x \forall y \text{ Vehicle}(x) \wedge \text{Vehicle}(y) \wedge \text{EmergencyType}(x) \wedge \text{EmergencyType}(y) \wedge \text{Conflict}(x,y) \wedge \text{Priority}(x,y) \rightarrow \text{ClearPath}(x) \wedge \neg \text{ClearPath}(y)$
- A temporal/queueing fact so the lower-priority vehicle is granted ClearPath only after the higher-priority one has passed, rather than being denied indefinitely.

With these additions the KB can logically resolve which of the two competing emergency vehicles gets the green signal first, while still eventually granting passage to both.

Case Study 3 — Agricultural Expert Reasoning System

(a) Conversion of P1, P2, P3 and Facts to CNF

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

Step 1 (Eliminate $\rightarrow$): $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

Step 2 ($\neg$ already innermost, nothing to distribute)

CNF: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

Step 1 (Eliminate $\rightarrow$): $\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

Step 2 (Move $\neg$ inward, De Morgan): $(\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat}) \vee \text{ApplyDripMethod}$

Step 3 (Distribute / flatten disjunction): $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

CNF: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Step 1 (Eliminate $\rightarrow$): $\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Step 2 (Move $\neg$ inward, De Morgan): $(\text{ApplyDripMethod} \vee \neg \text{CropWheat}) \vee \text{CropAtRisk}$

Step 3 (Distribute / flatten disjunction): $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

CNF: $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Facts

$\text{SoilDry} = \text{True} \rightarrow$ CNF unit clause: SoilDry

$\text{CropWheat} = \text{True} \rightarrow$ CNF unit clause: CropWheat

(b) Resolution Refutation — Prove 'ApplyDripMethod'

Negate the goal and add it to the CNF knowledge base:

Clauses:

C1: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

C2: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

C3: $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

C4: SoilDry

C5: CropWheat

C6 (negated goal): $\neg \text{ApplyDripMethod}$

#	Resolve	Resulting Clause
1	C1 with C4 (SoilDry)	IrrigationNeeded
2	Result-1 with C2	$\neg$ CropWheat $\vee$ ApplyDripMethod
3	Result-2 with C5 (CropWheat)	ApplyDripMethod
4	Result-3 with C6 ($\neg$ ApplyDripMethod)	$\square$ (empty clause)

Resolution Proof Tree:

$$\begin{array}{c}
 \text{C1}(\neg\text{SoilDry} \vee \text{IrrigationNeeded}) \quad \text{C4}(\text{SoilDry}) \\
 \quad \backslash \quad / \\
 \quad \text{IrrigationNeeded} \\
 \text{C2}(\neg\text{IrrigationNeeded} \vee \neg\text{CropWheat} \vee \text{ApplyDripMethod}) \\
 \quad \backslash \quad / \\
 \quad (\neg\text{CropWheat} \vee \text{ApplyDripMethod}) \quad \text{C5}(\text{CropWheat}) \\
 \quad \quad \backslash \quad / \\
 \quad \quad \text{ApplyDripMethod} \quad \text{C6}(\neg\text{ApplyDripMethod}) \\
 \quad \quad \quad \backslash \quad / \\
 \quad \quad \quad \square \text{ (contradiction)}
 \end{array}$$

Since the empty clause $\square$ is derived, the theorem ApplyDripMethod is proved TRUE by resolution refutation.

(c) Effect of Removing the Fact 'SoilDry'

Step-by-step re-evaluation:

- Without SoilDry as a fact, clause C1 ($\neg$ SoilDry $\vee$ IrrigationNeeded) can no longer be resolved with a unit clause for SoilDry, so IrrigationNeeded can no longer be derived.
- Since IrrigationNeeded is not established, clause C2 cannot fire, so ApplyDripMethod can no longer be derived from P2 either.
- Now examine P3: $\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk. CropWheat is still True (fact). ApplyDripMethod, however, is now simply UNKNOWN under classical (open-world) logic — it is not proven false, only unproven.
- Under strict classical/monotonic resolution, $\neg$ ApplyDripMethod is NOT a derivable literal (we only know it is not proved true), so P3's premise $\neg$ ApplyDripMethod $\wedge$ CropWheat cannot be confirmed, and CropAtRisk CANNOT be formally proved.

- If instead the system applies the Closed-World Assumption (CWA) / negation-as-failure — common in practical rule-based expert systems — anything not proven true is treated as false. Under CWA, ApplyDripMethod is treated as False, so $\neg$ ApplyDripMethod becomes True, and combined with CropWheat = True, P3 fires and CropAtRisk IS concluded.

Conclusion: Whether 'CropAtRisk' still follows depends on the reasoning pattern used — it does NOT follow under strict monotonic classical resolution (ApplyDripMethod is merely unknown, not false), but it DOES follow under the Closed-World Assumption / default reasoning typically used by practical expert systems. This distinction between classical negation and negation-as-failure is essential to a correct analysis of this KB.

Case Study 4 — Wumpus World Knowledge Agent

(a) Belief State Construction via Modus Ponens

Percepts received: Stench[1,2], Breeze[1,1], Glitter[2,2].

Modus Ponens form: from premises 'P' and 'P $\rightarrow$ Q', derive 'Q'.

Percept (P)	Rule Applied (P $\rightarrow$ Q)	Derived Fact (Q) via Modus Ponens
Stench[1,2]	Stench[1,2] $\rightarrow$ Wumpus adjacent to [1,2]	WumpusAdjacent([1,2]) — i.e. Wumpus is in one of the cells neighboring [1,2]
Breeze[1,1]	Breeze[1,1] $\rightarrow$ Pit adjacent to [1,1]	PitAdjacent([1,1]) — i.e. a Pit is in one of the cells neighboring [1,1]
Glitter[2,2]	Glitter[2,2] $\rightarrow$ Gold at [2,2]	Gold([2,2])

Belief state after applying Modus Ponens: {WumpusAdjacent([1,2]), PitAdjacent([1,1]), Gold([2,2])}.

(b) Forward Chaining to Locate the Wumpus and the Pit

Assume the standard grid-adjacency (cells sharing an edge are neighbors). Neighbors of [1,2] = {[1,1], [1,3], [2,2]}. Neighbors of [1,1] = {[1,2], [2,1]}.

Step	Fact in Working Memory	Reasoning	New Conclusion
1	WumpusAdjacent([1,2])	Wumpus $\in$ {[1,1],[1,3],[2,2]}	Candidate Wumpus cells listed
2	[1,1] is the agent's start cell (safe by convention)	Eliminate [1,1] from Wumpus candidates	Wumpus $\in$ {[1,3],[2,2]}
3	PitAdjacent([1,1])	Pit $\in$ {[1,2],[2,1]}	Candidate Pit cells listed
4	Agent is standing in [1,2] and perceiving normally (no fall)	[1,2] is not a Pit $\rightarrow$ eliminate [1,2]	Pit is at [2,1]
5	Rule 4: $\neg$ Wumpus[x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]	[1,1] and [1,2] have neither confirmed $\rightarrow$ Safe([1,1]), Safe([1,2])	Safe cells: {[1,1],[1,2]}

Result: Pit is confidently located at [2,1]. The Wumpus is narrowed to {[1,3], [2,2]} — a single additional percept check at [1,3] (absence or presence of stench) would fully disambiguate its location.

(c) Safety Evaluation of Path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$

- $[1,1]$: The agent's starting square is safe by the standard Wumpus World convention, and Rule 4 confirms $\text{Safe}([1,1])$ since neither Wumpus nor Pit is asserted there.
- $[1,2]$: The agent has already visited this cell to perceive the Stench, and survived, confirming $\neg\text{Pit}([1,2])$. The Stench rule only places the Wumpus adjacent to $[1,2]$, not inside it, so $\neg\text{Wumpus}([1,2])$ also holds. By Rule 4, $\text{Safe}([1,2])$ is confirmed.
- $[2,2]$: This cell holds the Gold (from Glitter), but it is also one of the two remaining unresolved candidate cells for the Wumpus (from part b, $\text{Wumpus} \in \{[1,3], [2,2]\}$). Since $\neg\text{Wumpus}([2,2])$ has NOT been established, Rule 4's premise $\neg\text{Wumpus}[x,y] \wedge \neg\text{Pit}[x,y]$ cannot be confirmed for $[2,2]$, so $\text{Safe}([2,2])$ cannot be logically derived.

Conclusion: The path $[1,1] \rightarrow [1,2]$ is provably safe, but proceeding directly into $[2,2]$ to collect the gold is NOT logically justified with the current knowledge — there remains a live possibility that the Wumpus occupies $[2,2]$. A rational logical agent should first visit $[1,3]$ (a cell not adjacent to any risk yet identified) and check for Stench there: if $[1,3]$ shows no Stench, the Wumpus is eliminated from $[1,3]$ and, since it must be at $[1,3]$ or $[2,2]$, is thereby confirmed at $[2,2]$ — meaning the agent should NOT enter $[2,2]$. If $[1,3]$ does show Stench, the Wumpus is confirmed at $[1,3]$ instead, and $[2,2]$ becomes provably safe ($\neg\text{Wumpus}([2,2])$ and $\neg\text{Pit}([2,2])$ both hold), after which the gold may be safely collected.